

DIETING SYSTEM BASED ON DISEASE

AIM

To write a Prolog program that suggests a suitable diet based on a person's disease using facts and rules.

ALGORITHM

1. Start.
2. Define diseases and their recommended foods.
3. Define foods to avoid.
4. Accept a disease as a query.
5. Search the database for suitable food.
6. Display the recommended diet.
7. Stop.

SOURCE CODE

% Diet Recommendation System

diet(diabetes, vegetables).

diet(diabetes, whole_grains).

diet(diabetes, salads).

diet(hypertension, fruits).

diet(hypertension, vegetables).

diet(hypertension, low_salt_food).

diet(obesity, vegetables).

diet(obesity, fruits).

diet(obesity, low_calorie_food).

avoid(diabetes, sugary_food).

avoid(hypertension, salty_food).

avoid(obesity, fast_food).

recommend(Disease, Food) :-

diet(Disease, Food).

OUTPUT

```
% c:/Users/Abinaya Sri J/OneDrive/Desktop/Prolog exercise/DIET.pl
?- recommend(diabetes, Food).
Food = vegetables ;
Food = whole_grains ;
Food = salads.

?- recommend(hypertension, Food).
Food = fruits ;
Food = vegetables ;
Food = low_salt_food.

?-
```

RESULT

The Prolog program successfully suggests diet items based on the given disease using a simple rule-based knowledge base.